

CS101

Computer Hardware

1/20

Agenda

1. Computer Hardware Video+Lecture
2. Operating Systems Articles
3. Windows OS Tips (time permitting)
4. Work on Lab1/Codecademy
5. Touch Typing Initial Test

10 min break at ~5pm

Computer Hardware

<https://www.youtube.com/watch?v=HB4I2CgkcCo>

Central Processing Unit (CPU)



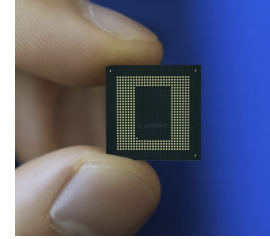
- CPUs work serially
 - Multitask by rapidly switching between different tasks
- Clock Speed: processing cycles (ticks) per second.
 - 1 GHz: 1 Billion ticks per second
- 1CPU Can have multiple “Cores”

CPU Examples

Computer



Smartphone



- Modern CPUs all 4-40 Cores, 3-4 GHz
 - Laptops
 - **Intel** i7 (Windows/Chromebook)
 - **AMD** Ryzen (Windows)
 - **Apple** M4 (Macbook)
 - Phones
 - **Qualcomm** Snapdragon (Samsung)
 - **Google** Tensor (Google Pixel)
 - **Apple** A19 (iPhone)

Bits and Bytes

Units of Digital Data Storage

UNIT	SYMBOL	EQUIVALENCE	IN BYTES	REFERENCE
Bit	b	Binary digit (0 or 1)	$\frac{1}{8}$ Byte	A single 0 or 1
Byte	B	8 bits	1 Byte	One letter
Kilobyte	KB	1,024 bytes	~Thousand (10^3) Bytes	~20 lines of a book
Megabyte	MB	1,024 KB	~Million (10^6) Bytes	1 minute of a song
Gigabyte	GB	1,024 MB	~Billion (10^9) Bytes	30 minutes of HD video
Terabyte	TB	1,024 GB	~Trillion (10^{12}) Bytes	200 HD movies

Hard Drive



Storage 128 GB → 20+ TB

Speed 80-200 MB/s

Solid State Hard Drive (SSD)



Same Storage Ranges

Speed 500 MB/s → 7 GB/s

Random Access Memory (RAM)

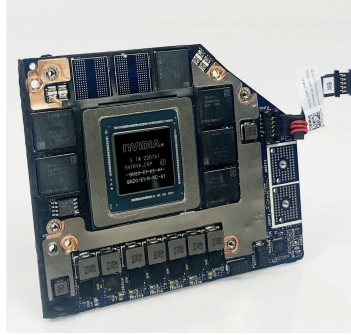


Storage 4-128GB (Need at least 8GB for programming)

Speed 3200 MHz → 25 GB/s

Newer → 60 GB/s

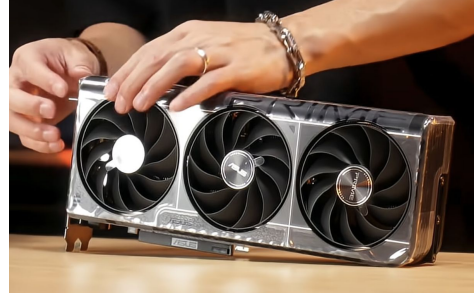
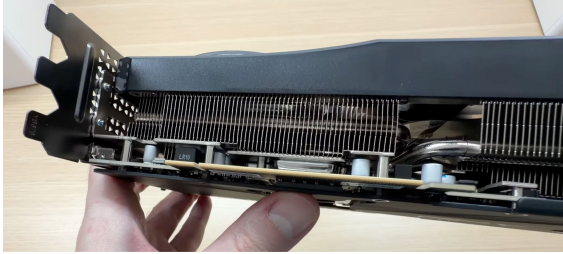
Graphics Processing Unit (GPU)



- 1,000-16,000 cores
 - Simple operations in parallel
 - GPU core slower than CPU core
- Tasks:
 - Updating video game scene
 - Training AI Models
- Has own RAM

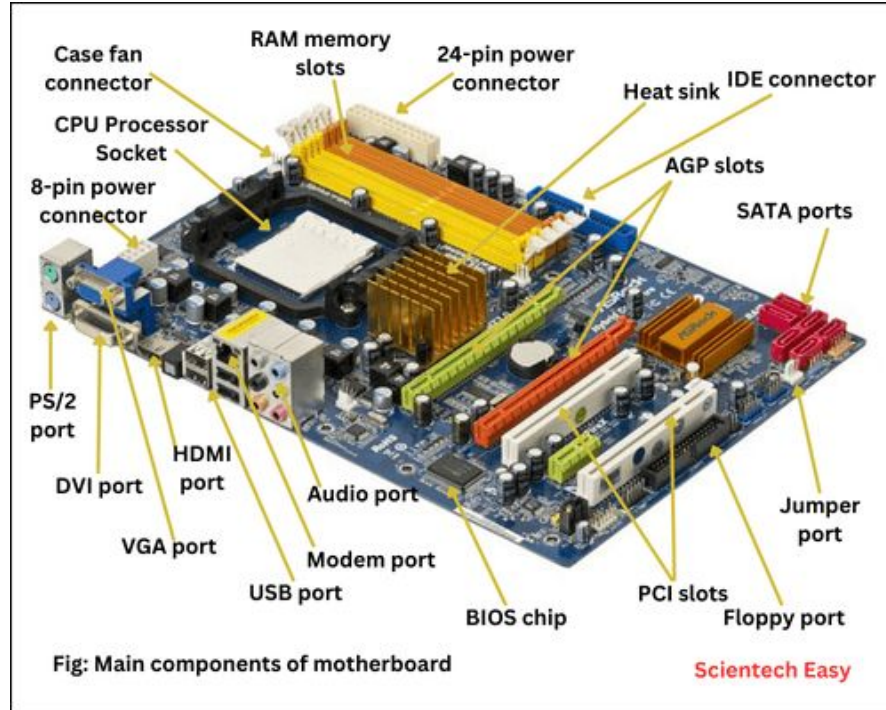
GPU Examples

Modern Gaming/AI GPUs



- New Chips integrate GPUs with CPUs

Motherboard connects the pieces



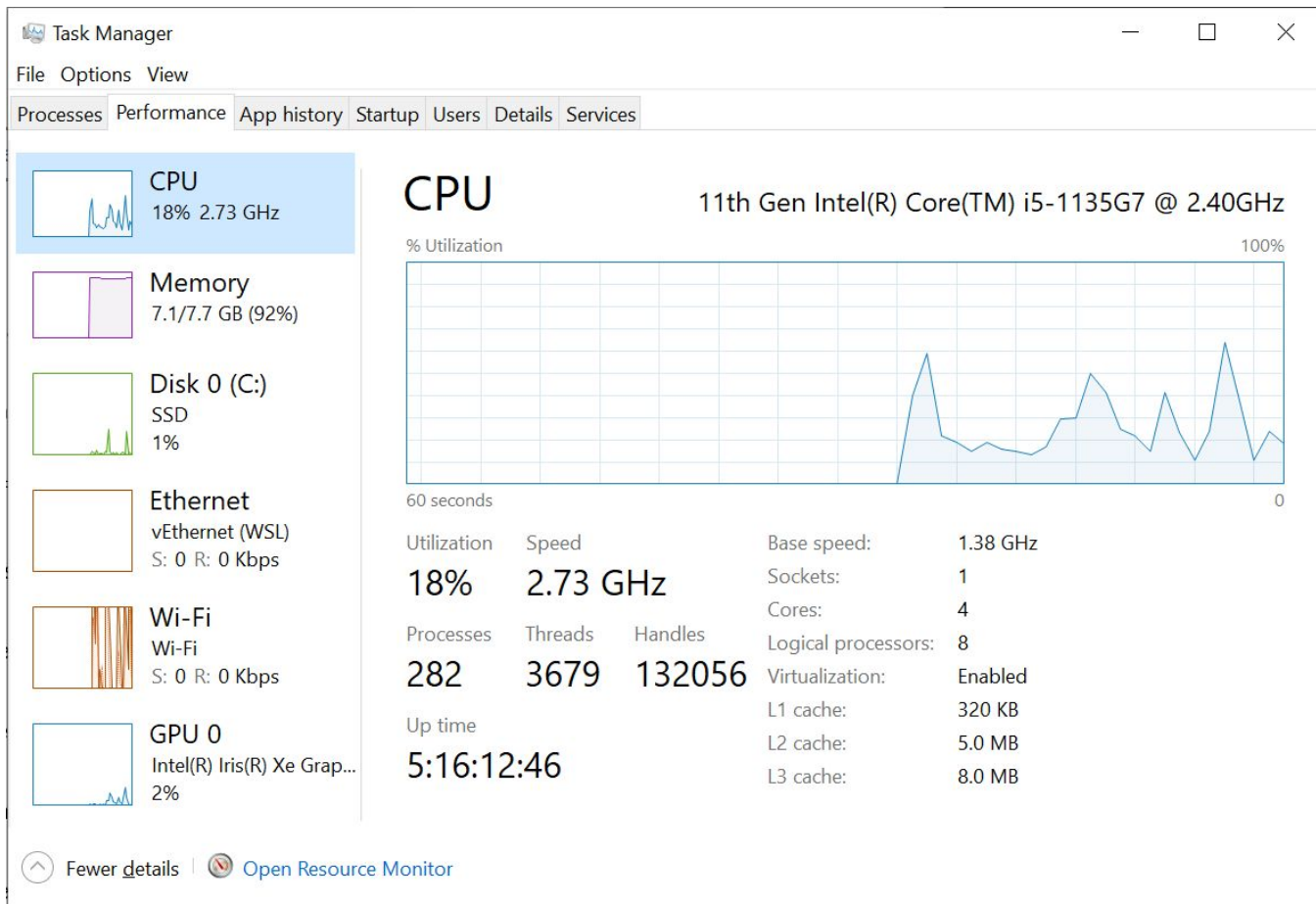
Data Centers and GPU Centers



Data Center $\rightarrow$ 1EB Storage = $\sim 10^6$ TB

GPU Center $\rightarrow$ 15,000 GPUs

Task Manager on Windows



Operating Systems

Read handouts:

- <https://medium.com/@tanalpha-aditya/introduction-to-operating-systems-and-networks-dfa3611befcc>
- <https://www.geeksforgeeks.org/operating-systems/introduction-of-operating-system-set-1/>

Operating Systems Comparison

Operating System	Kernel	CLI	GUI
Windows (Microsoft)	NT Kernel	PowerShell	Windows Shell (Explorer)
macOS (Apple)	Unix	Bash / Zsh	Aqua
Linux (Open Source)	Unix	Bash / Zsh	GNOME / KDE / others
Chrome OS (Google)	Unix	Bash / Crosh	Aura

Unix preferred for programming

Next Class 1/22

1. Quiz 1
2. See inside a Macbook
3. Lecture
4. Start Lab2

Everyone **MUST** come to an office hours or schedule a 1on1 **before 1/29**. (Food reward if all succeed)